

Maximum likelihood estimation

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Logistics and reminders

- + HW 1 due tonight
- + HW 2 released, due next Friday
- + Project part 1 released (choosing a dataset + EDA)
 - + Project will be released in several pieces
 - + Part 1 due in 2 weeks
- + Department seminar coming up on 9/11
 - + 11am in ZSR auditorium
 - + Speaker: Robert Langefeld
 - + Attendance part of class participation grade
 - + If you can't attend in person, can instead watch a seminar on YouTube

Recap

Definition: The *likelihood* $L(Model) = P(Data|Model)$ of a model is the probability of the observed data, given that we assume a certain model and certain values for the parameters that define that model.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Model: $Y_i \sim Bernoulli(\pi)$, and $\hat{\pi} = 0.9$
- + Data: $y_1, \dots, y_5 = T, T, T, T, H$
- + Likelihood: $L(\hat{\pi}) = P(y_1, \dots, y_5 | \pi = 0.9) = 0.00009$

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

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How do I maximize functions?

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Step 1: Write down the likelihood

$$L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Step 2: Take the log

$$\log L(\hat{\pi}) =$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Step 3: maximize

$$\log L(\hat{\pi}) = 4 \log(1 - \hat{\pi}) + \log(\hat{\pi})$$

How do we find where maxima/minima occur for a function?

Maximum likelihood estimation with calculus

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\hat{\pi}) = (1 - \hat{\pi})^4(\hat{\pi})$
- + Choose $\hat{\pi}$ to maximize $L(\hat{\pi})$

Step 3: maximize -- take first derivative and set equal to 0

$$\log L(\hat{\pi}) = 4 \log(1 - \hat{\pi}) + \log(\hat{\pi})$$

$$\frac{d}{d\hat{\pi}} \log L(\hat{\pi}) =$$

Class activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class activity

+ $P(Y_i = 0) = \pi_0$

+ $P(Y_i = -1) = 2\pi_0$

+ $P(Y_i = 1) = 1 - 3\pi_0$

Observe data $-1, -1, 0, 1, 0, -1$.

$$L(\hat{\pi}_0) = ?$$

Class activity

$$\log L(\hat{\pi}_0) = 3 \log(2) + 3 \log(\hat{\pi}_0) + 2 \log(\hat{\pi}_0) + \log(1 - 3\hat{\pi}_0)$$

$$\frac{d}{d\hat{\pi}_0} \log L(\hat{\pi}_0) =$$

Maximum likelihood in more generality

Suppose that $Y_i \sim \text{Bernoulli}(\pi)$. We observe n observations $Y_1, \dots, Y_n$ and want to estimate π .

Step 1: Write down the likelihood

- + Let $\hat{\pi}$ be the estimate of π
- + Let k be the number of times $Y_i = 1$ in the data

$$L(\hat{\pi}) =$$

Maximum likelihood in more generality

Step 1: Write down the likelihood

$$L(\hat{\pi}) = \hat{\pi}^k (1 - \hat{\pi})^{n-k}$$

Step 2: Take the log

$$\log L(\hat{\pi}) =$$

Maximum likelihood in more generality

Step 3: take the first derivative, and set = 0

$$\log L(\hat{\pi}) = k \log(\hat{\pi}) + (n - k) \log(1 - \hat{\pi})$$

$$\frac{d}{d\hat{\pi}} \log L(\hat{\pi}) =$$

Maximum likelihood in more generality

So our maximum likelihood estimate is $\hat{\pi} = \frac{k}{n}$, the sample proportion

+ Example data: T, T, T, T, H

+ This implies that $\hat{\pi} = \frac{1}{5} = 0.2$